

END SEMESTER EXAMINATION, 2023-24**Set A****(TERM : 2302)****ARP 208 : QUANTITATIVE AND QUALITATIVE APTITUDE SKILL BUILDING****Time : 2 Hrs.****Max. Marks: 100**

Note: 1. All Sections are compulsory.
2. Assume missing data suitably, if any.

- CO1: Develop and deliver effective presentations to interpret the deeper meaning of life.
CO2: Improve listening skills to understand complex business communication in a variety of global English accents through proper pronunciation
CO3: Demonstrate a good understanding of effective business writing and telephone handling Skills
CO4: Acquire higher level competency in the use of aptitude, logical and analytical reasoning
CO5: Develop higher-level strategic thinking and diverse mathematical concepts through building number puzzles
CO6: Demonstrate higher level quantitative aptitude tools for making business decisions.

SECTION-A**VARBAL ABILITY****Attempt ALL questions of the following:****(50 × 1 = 50)**

Directions for Questions 1 – 5: Read the passage and answer the questions that follow **CO_s BTL**

RC1 I grew up in a small town not far from Kalimpong. In pre-liberalization India, everything arrived late; not just material things but also ideas. Magazines-old copies of Reader's Digest and National Geographic-arrived late too, after the news had become stale by months or, often, years. This temporal gap turned journalism into literature, news into legend, and historical events into something akin to plotless stories. But like those who knew no other life, we accepted this as the norm. The dearth of reading material in towns and villages in socialist India is hard to imagine, and it produced two categories of people: those who stopped reading after school or college, and those including children who read anything they could find. I read road signs with the enthusiasm that attaches to reading thrillers. When the itinerant kabadiwala, collector of papers, magazines, and rejected things, visited our neighborhood, I rushed to the house where he was doing business. He bought things at unimaginably low prices from those who'd stopped having any use for them, and I rummaged through his sacks of old magazines. Sometimes, on days when business was good, he allowed me a couple of copies of Sportsworld magazine for free. I'd run home and, ignoring my mother's scolding, plunge right into the in-consuming news about India's victory in the Benson and Hedges Cup. **CO4 K4**

Two takeaways from these experiences have marked my understanding of the provincial reader's life: the sense of belatedness, of everything coming late, and the desire for pleasure in language..... Speaking of belatedness, the awareness of having been born at the wrong time in history, of inventing things that had already been discovered elsewhere, far away, without our knowledge or cooperation, is a moment of epiphany and deep sadness. I

Remember a professor's choked voice, narrating to me how all the arguments he'd made in his doctoral dissertation, written over many, many years of hard work (for there indeed was a time when PhDs were written over decades), had suddenly come to naught after he'd discovered the work of C.W.E. Bigsby. This, I realized as I grew older, was one of the characteristics of provincial life: that they (usually males) were saying trite things with the confidence of someone declaring them for the first time. I, therefore, grew up surrounded by would-be Newtons who claimed to have discovered gravity (again). There's a deep sense of tragedy attending this sort of thing-the sad embarrassment of always arriving after the party is over. And there's a harsh word for that sense of belatedness: "dated." What rescues it is the unpredictability of these anachronistic "discoveries" - the randomness and haphazardness involved in mapping connections among thoughts and ideas, in a way that hasn't yet been professionalized.

- 1 What use was the kabadiwala (waste picker) to the author? CO4 K4
 - (a) The kabadiwala bought up all her magazines
 - (b) The kabadiwala's stock of books and magazines were of interest to the author
 - (c) The kabadiwala was about to steal the author's magazines
 - (d) The author ordered books online which the kabadiwala delivered
- 2 What according to the author is essential about the experience of being a 'provincial reader'? CO4 K4
 - (a) Belatedness in the sense of coming late for everything.
 - (b) Over-eagerness.
 - (c) Accepting a temporal gap between what was current in the wider world and the time at which these arrived in the provincial location.
 - (d) None of the above
- 3 Why did the author feel a sense of epiphany and deep sadness? CO4 K4
 - (a) Because the things that felt special and unique to the author, were already established and accepted in the wider world.
 - (b) Because the author was less well-read than others.
 - (c) Because the author missed being in a big city.
 - (d) All the above
- 4 What does the word 'anachronistic' as used in the passage, mean? CO4 K4
 - (a) Rooted in a non-urban setting
 - (b) Related to a mofussil area
 - (c) Connected with another time
 - (d) Opposed to prevailing sensibilities
- 5 Which of the following options captures the meaning of the last sentence best? CO4 K4
 - (a) Though the author feels provincial, she pretends to be from the metropolis.
 - (b) Though the author feels dated in her access to intellectual ideas, her lack of metropolitan sophistication lets her engage with the ideas with some originality.
 - (c) Though the author is aware of the limitedness of her knowledge, she is confident and can hold her own in a crowd. She also proud of her roots in the small town.
 - (d) All the above

Directions for Questions 6 – 10: Read the passage and answer the questions that follow **CO_s** **BTL**

- RC2** Some people say you are the kind of person you are because of when you were born in your family. They say, for example, being born first makes you more responsible. A number of researchers say, where you fit in your family has a big influence on how you will act, how well you do in school and how much money you'll make. They say first-borns earn the most. Some researchers, say birth order differences are as strong as gender difference. 'Within the family, they are about as strong as gender differences," according to Frank Sulloway, "later-borns tend to rebel because they often can't do what their older siblings can do, so they start trying to find other ways, even dangerous ones, to get their parents' attention". Sulloway points out that leaders of revolutions --- like Thomas Jefferson, Karl Marx and 'Fidel Castro --- were rebellious younger brothers and older brothers are often more conservative like former. Presidents Carter and Clinton than their younger brothers; Billy and Roger. Billy Carter had a beermaking business, and Roger Clinton 'tried a singing career -- a far cry from presidential politics. Sulloway says that younger siblings tend to pick interests that are diametrically opposite to those of their older siblings, that they're the risk takers, the adventures, the people who are constantly trying to find something new and different to do. Sulloway also says that later-borns rebel by choosing different professions. "Later-borns rebel", says Sulloway, "because they're controlled by the first-born. The typical first-born strategy is to use the advantages of age, size and power to dominate a younger sibling", Subway says.
- 6 Later-borns act differently in order to CO4 K4
 (a) get as many opportunities as possible. (b) claim their parents' attention.
 (c) compete with their eldest siblings. (d) be more successful.
- 7 Later--borns rebel by CO4 K4
 (a) taking risk.
 (b) choosing professions different from those of their elder siblings,
 (c) trying to find different ways. (d) All of the above
- 8 Which of the following has NOT been stated in the passage to beta a difference seen between the first-born and the later-borns? CO4 K4
 (a) The first-born is more responsible. (b) The firstborn is more conservative.
 (c) Later-borns are more amiable. (d) Later-borns are more adventurous.
- 9 Later-borns rebel because CO4 K4
 (a) first-borns use the advantage of age to control them.
 (b) their interests are diametrically opposite to those of their older siblings.
 (c) first-borns are too conservative.
 (d) first-bores are highly successful.
- 10 The actions, educational performance and earnings of sibling according to the passage, are influenced by CO4 K4
 (a) gender difference (b) parental affection.
 (c) birth order (d) the first-born.

Directions for Questions 11–15: Read the passage and answer the questions that follow **CO_s** **BTL**

- RC3** How do other people perceive you, especially upon first meeting you face-to-face? How well do you anticipate another person's discomfort before that person freezes up and becomes paralyzed, withdrawn or even destructive in a situation? Whichever side of the table you are on, these skills are crucial to your ability to lead, mentor or be a valuable team player with your staff, vendors and

customers. Most people cannot help 'leaking' their feelings. Fortunately, few of us are attuned to noticing the often subtle signals that indicate strong emotions in others. Or we misread the signals. Your body is a hologram of your being; a three-dimensional move that is constantly running, showing others how you feel about yourself and the worlds. As you work through life, is your body saying what your words are saying? Your body is a three-dimensional "full-motion" billboard to the rest of the world. Your body speaks to you all of the time, telling you your own needs. Listen to it. It is your free and most sophisticated medical feedback testing system, continuously showing you your inner tensions, state of mind, and habitual life attitudes. When you are misaligned and tense, you expend outrageous sums of extra energy in the everyday gestures of life. Because the body is a high-viscosity substance that is 60-80% water, your bones are floating in a relatively fluid environment. Over a period of time, despite that apparent fluidity, you have tightened the muscles around every major experience of pain, fear, or anger. In Western society, people usually hold the tension somewhere in their upper bodies, whereas in many Eastern cultures, the tension tends to be held in the lower body. We go through life-making decisions, closing down and limiting ourselves unconsciously. If you don't begin a regular practice of exercise and stretching, you are guaranteed to lose mobility sooner as you age, robbing yourself of the most positive and alive present you can offer the world every day - a loose and relaxed presence.

Stay open literally by getting in motion more frequently. Stand and stretch at least every twenty minutes when you are sitting and working. Try to walk, hopefully in sync with someone else, in fresh air and sunlight, at least thirty minutes a day. As Dr. Dean Ornish wrote in his most recent book, *Love and Survival: The Scientific Basis for the Healing power of Intimacy*, our survival depends on the healing power of love. One of the safest and most natural ways to move closer to others is to walk with them. Walk farther to the restaurant. Walk and talk on the way to the meeting. Walk with your loved one, rather than sitting at home, to come down from your day together. Motion is emotional and makes every event more vivid and memorable. Literally move toward the one you want in your life and loosen up together. Your life could depend on it

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|----|---|-----|----|
| 11 | <p>'These skills' mentioned in the last sentence of the first para refer to</p> <p>(a) being perceptive to others and being able to present a picture of yourself that you want others to see.</p> <p>(b) conducting face-to-face meetings and convincing others.</p> <p>(c) anticipating how others become paralyzed, withdrawn or even destructive in a situation and then helping them.</p> <p>(d) anticipating how others who feel uncomfortable react and making a note of it.</p> | CO4 | K4 |
| 12 | <p>It is fortunate, according to the author, that few of us are attuned to noticing the often subtle signals because,</p> <p>(a) they indicate strong emotions.</p> <p>(b) we misuse the signals.</p> <p>(c) most people are helpless in expressing their feelings.</p> <p>(d) we would most probably feel uncomfortable if they knew how we feel.</p> | CO4 | K4 |
| 13 | <p>It can be understood that it is desirable that</p> <p>(a) we show the world how we feel.</p> <p>(b) we evaluate our body language and stop responding.</p> <p>(c) we match what we say and what signals we are sending.</p> <p>(d) we also consciously react to those we are communicating with</p> | CO4 | K4 |

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|----|--|-----|----|
| 14 | The most sophisticated medical feedback testing system, according to the passage, is | CO4 | K4 |
| | (a) our body signals. (b) our body | | |
| | (c) our perception (d) All of the above | | |
| 15 | The tension, generated by strong emotions in can be loosened by | CO4 | K4 |
| | (a) regular practice of exercise and stretching. | | |
| | (b) standing and stretching every twenty minutes when at work. | | |
| | (c) taking walks with others thus moving closer to them. | | |
| | (d) All of the above. | | |

Directions for Questions 16 – 20: Read the passage and answer the questions that follow: CO_s BTL

RC4 Ragtime is musical form that synthesizes folk melodies and musical techniques CO4 K4

into a brief quadrille-like structure, designed to be played – exactly as written – on the piano. A strong analogy exists between European composers like Ralph Vaughan Williams, Edward Grieg, and Anton Dvorak who combined folk tunes and their own original materials in larger compositions and the pioneer ragtime composers in the United States. Composers like Scott Joplin and James Scott were in a sense collectors or musicologists, collecting dance and folk music in Black communities and consciously

shaping it into brief suites or anthologies called piano rags. It has sometimes been charged that ragtime is mechanical. For instance, Wilfred Mellers comments, “rags were transferred to the piano roll and, even if not played by a machine, should be played like a machine, with meticulous precision.” However, there is no reason to assume that ragtime is inherently mechanical simply because commercial manufacturers applied a mechanical recording method to ragtime, the only way to record pianos at that date. Ragtime’s is not a mechanical precision, and it is not precision limited to the style of performance. It arises from ragtime’s following a well-defined form and obeying simple rules within that form. The classic formula for the piano rag disposes three to five themes in sixteen-bar strains, often organized with repeats. The rag opens with a bright, memorable strain or theme, followed by a similar theme, leading to a trio of marked lyrical character, with the structure concluded by a lyrical strain that parallels the rhythmic developments of the earlier themes.

The aim of the structure is to rise from one theme to another in a stair step manner, ending on a note of triumph or exhilaration. Typically, each strain is divided into two 8-bar segments that are essentially alike, so the rhythmic-melodic unit of ragtime is only eight bars of 2/4 measure. Therefore, themes must be brief with clear, sharp melodic figures. Not concerned with development of musical themes, the ragtime composer instead sets a theme down intact, in finished form, and links it to various related themes. Tension in ragtime compositions arises from a polarity between two basic ingredients: a continuous bass—called by jazz musicians a boom chick bass—in the pianist’s left hand, and its melodic, syncopated counterpart in the right hand. Ragtime remains distinct from jazz both as an instrumental style and as a genre. Ragtime style stresses a pattern of repeated rhythms, not the constant inventions and variations of jazz. As a genre, ragtime requires strict attention to structure, not inventiveness or virtuosity. It exists as a tradition, a set of conventions, a body of written scores, separate from the individual players associated with it. In this sense ragtime is more akin to folk music of the nineteenth century than to jazz.

- 16 Which of the following best describes the main purpose of the passage? CO4 K4
- To contrast ragtime music and jazz.
 - To acknowledge and counter significant adverse criticisms of ragtime music.
 - To define ragtime music as an art form and describe its structural characteristics.
 - To review the history of ragtime music and analyse ragtime's effect on listeners.
- 17 According to the passage, each of the following is a characteristic of ragtime compositions that follow the classic ragtime formula except CO4 K4
- well-defined melodic figures.
 - rising rhythmic-melodic intensity.
 - a bass line distinct from the melodic line.
 - full development of musical themes.
- 18 According to the passage, Ralph Vaughan Williams, Anton Dvorak, and Scott Joplin are similar in that they all CO4 K4
- conducted research into musicological history.
 - wrote original compositions based on folk tunes.
 - collected and recorded abbreviated piano suites.
 - created intricate sonata-like musical structures.
- 19 The author rejects the argument that ragtime is a mechanical music because that argument CO4 K4
- overlooks the precision required of the ragtime player.
 - does not accurately describe the sound of ragtime piano music.
 - confuses the means of recording and the essential character of the music.
 - exaggerates the influence of the performance style of professional ragtime players on the reputation of the genre.
- 20 It can be inferred that the author of the passage believes that the most important feature of ragtime music is its: CO4 K4
- Commercial success
 - Formal structure
 - Emotional range
 - Improvisational opportunities
- Directions for Questions 21 - 35: Fill in the blank space(s) with the right combination of word(s)** COs BTL
- 21 Restlessness among the students of colleges and universities has _____ from their deep rooted feeling that their views and aspirations are _____ by their elders. CO2 K2
- arisen, humiliated
 - followed, disregarded
 - stemmed, ignored
 - started, neglected
- 22 Shalini was not _____ by the criticism and paid no _____ even when her best friend talked against her. CO2 K2
- bothered, attention
 - troubled, mind
 - threatened, warning
 - deterred, heed
- 23 He granted the request because he was _____ to _____ his friend. CO2 K2
- Reluctant, disappoint
 - Unwilling, please
 - Bound, hurt
 - Destined, agonise

- 24 The charges made in the system were so _____ that they didn't require any _____. CO2 K2
 (a) Obvious, modifications (b) Certain, expertise
 (c) Big, time (d) Genuine, intelligence
- 25 In _____ of International matters, there is always an element of risk in _____ one might do. CO2 K2
 (a) Defence, wrong (b) Case, whatever
 (c) View, whichever (d) Many, doing
- 26 _____ is a criminal _____ in England and covers cases where offensive descriptions of Christianity are published. CO2 K2
 (a) Blasphemy, offence (b) Obscenity, deviation
 (c) Heathenism, act (d) Sacrilege, violence
- 27 We are _____ to have him _____ here to make this function a great success. CO2 K2
 (a) Sure, come (b) Pleased, over (c) Proud, leave (d) Happy, arrive
- 28 He was so convinced that people were driven by _____ motives that he believed there was no such thing as a purely _____ act. CO2 K2
 (a) Altruistic, praiseworthy (b) Ulterior, selfless
 (c) Personal, anti-social (d) Personal, eternal
- 29 We _____ him with many promises, but nothing would _____ him. CO2 K2
 (a) Attracted, fascinate (b) Gave, deprive
 (c) Tempted, influence (d) Provoked, dessicate
- 30 The counter clerk was very busy and _____ not pay _____ to Sameer's request CO2 K2
 (a) Had, cash (b) Did, attention
 (c) Can, help (d) Could, respect
- 31 The construction of the ball has been _____ because of the _____ of cement in the market. CO2 K2
 (a) Hampered, shortage (b) Prevented, supply
 (c) Held, non-availability (d) Denied, restrictions
- 32 The bandit _____ the traveller of his purse of gold and _____ him grievously. CO2 K2
 (a) Demanded, beat (b) Robbed, wounded
 (c) Snatched, hurt (d) Stole, injured
- 33 She was _____ because all her plans had gone _____. CO2 K2
 (a) distraught, awry (b) Frustrated, magnificently
 (c) Elated, wild (d) Dejected, splendidly
- 34 The candidate's _____ at the polls was _____ as he won with a striking margin. CO2 K2
 (a) Image, real (b) Strategy, unsuccessful
 (c) Claim, unrealistic (d) Victory, overwhelming
- 35 He is so _____ that everyone is always _____ to help him in his work. CO2 K2
 (a) Magnanimous, eager (b) Miserly, ignorant
 (c) Helpful, reluctant (d) Aloof, cooperative

Directions for Questions 36 - 40: Fill in the blank space(s) with the CO_s BTL
GRAMMATICALLY CORRECT SENTENCE

- 36 Manufacturers rate batteries in watt-hours; if they rate the watt-hour higher, the longer the battery can be expected to last. CO₆ K3
 (a) if they rate the watt-hour higher, the longer
 (b) rating the watt-hour higher, it is that much longer
 (c) the higher the watt-hour rating, the longer
 (d) the higher the watt-hour rating, it is that much longer that
- 37 In 1979 lack of rain reduced India's rice production to about 41 million tons, nearly 25 percent less than those of the 1978 harvest. CO₆ K3
 (a) less than those of the 1978 harvest
 (b) less than the 1978 harvest
 (c) less than 1978
 (d) fewer than 1978
- 38 Unlike the nests of leaf cutters and most other ants, situated underground or in pieces of wood, raider ants make a portable nest by entwining their long legs to form "curtains" of ants that hang from logs or boulders, providing protection for the queen and the colony larvae and pupae. CO₆ K3
 (a) the nests of leaf cutters and most other ants,
 (b) the nests of leaf cutters and most other ants, which are
 (c) leaf cutters and most other ants, whose nests are
 (d) leaf cutters and most other ants in having nests
- 39 Discussion of greenhouse effects have usually had as a focus the possibility of Earth growing warmer and to what extent it might, but climatologists have indicated all along that precipitation, storminess, and temperature extremes are likely to have the greatest impact on people. CO₆ K3
 (a) Discussion of greenhouse effects have usually had as a focus the possibility of Earth growing warmer and to what extent it might,
 (b) Discussion of greenhouse effects has usually had as its focus whether Earth would get warmer and what the extent would be,
 (c) Discussion of greenhouse effects has usually focused on whether Earth would grow warmer and to what extent,
 (d) The discussion of greenhouse effects have usually focused on the possibility of Earth getting warmer and to what extent it might,
- 40 With corn, soybean, and wheat reserves being low enough so a poor harvest would send prices skyrocketing, grain futures brokers and their clients are especially interested in weather that could affect crops. CO₆ K3
 (a) being low enough so
 (b) so low such that
 (c) so low that
 (d) that are low enough so

Directions for Questions 41 - 45: Choose the GRAMMATICALLY INCORRECT CO_s BTL
PORTION of the sentence

- 41 I and Ganesh (a) / will take care of (b) / the function (c) / on Sunday (d) CO₆ K4

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| 42 | The SP has declared that those who are found (a) / guilty of breach of law (b) / or taking the law into their hands (c) / they will be taken to task (d) | CO6 | K4 |
| 43 | Now a days (a) / the weather (b) /is getting more cold (c) / and colder (d) | CO6 | K4 |
| 44 | All the books (a) / were indeed interesting (b) / but that one was (c) / the more interesting (d) | CO6 | K4 |
| 45 | The number (a) / of students interested (b) / in psychology are (c) / generally small (d) | CO6 | K4 |

Directions for Questions 46 - 50: In the following items, each passage consists of six sentences. The first sentence S1 and the final sentence S6 are given in the beginning. The middle four sentences in each have been removed and jumbled up. These are labeled P. Q. R and S. You are required to find out the proper sequence of the four sentences.

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| 46 | <p>S1 : It was Saturday.</p> <p>S6 : The children had already bought a roasted gram and peanuts to get into a picnic mood.</p> <p>P : A taxi carried us all to the zoo.</p> <p>Q : They wanted to be taken out and we decided to take them to the local zoo.</p> <p>R : My sister's two young children were at our house.</p> <p>S : We bought the tickets and entered the zoo.</p> <p><i>Which one of the following is the correct sequence?</i></p> <p>(a) RQSP (b) RQPS (c) PQS d) PRQS</p> | CO3 | K3 |
| 47 | <p>S1 : The officer rose to his feet, trembling.</p> <p>S6 : A half-hour later he returned to camp.</p> <p>P : He failed to find him there.</p> <p>Q : Pulling himself together, he ran rapidly away from the cliff to a point a half-mile from its foot.</p> <p>R : He was disappointed.</p> <p>S : He expected to find the horseman somewhere there.</p> <p><i>Which one of the following is the correct sequence?</i></p> <p>(a) RPQS (b) QPSR (c) QSPR (d) SQPR</p> | CO3 | K3 |
| 48 | <p>S1 : It was a bitterly cold night and even at the far end of the bus the wind cut like a knife.</p> <p>S6 : I saw trouble brewing.</p> <p>P : The conductor came in and took the fares.</p> <p>Q : The younger of the two women was dressed in sealskin and carried one of those little Pekinese dogs that women like to carry in their laps.</p> <p>R : The bus stopped and two women and a man got in together and filled the vacant places.</p> <p>S : Then, his eyes tested with cold malice on the beady-eyed little dog.</p> <p><i>Which one of the following is the correct sequence?</i></p> <p>(a) RQPS (b) RSQP (c) RPQS (d) PSRQ</p> | CO3 | K3 |
| 49 | <p>S1 : Dinner had been served—his daughter laid out the plates.</p> <p>S6 : Then, silently she left the table to retire for the night—it was as if she had never been there.</p> <p>P : She was just a child, only 14 —too young, too simple to know to understand.</p> <p>Q : He sat clown groundly, not saying a word to her.</p> | CO3 | K3 |

R : She had already had her meal and was standing by his side, not quite knowing what to do.

S : Bread and cheese—a simple subsistence at the end of a not-so-simple life.

Which one of the following is the correct sequence?

- (a) QRPS (b) RPSQ (c) QPSR (d) SQPR

- 50 S1 : When my car broke down, I took it to the only mechanic available in our town. CO3 K3

S6 : They pushed the car down one street and up another and soon we had gone through most of the streets in the town but the car would not start.

P : But it just refused to start.

Q : I went there at the appointed time to collect it.

R : So, I sat at the wheel and the mechanic and his helper started to push it.

S : He said it required some minor repairs and asked me to collect it in the evening.

Which one of the following is the correct sequence?

- (a) SQPR (b) QSRP (c) PQRS (d) RPQS

SECTION-B

APTITUDE AND REASONING

Attempt **ALL** questions of the following:

(50 × 1 = 50)

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| | | CO_s | BTL |
| 51. | On Ashok Marg, three consecutive traffic lights change after 36, 42, and 72 seconds respectively. If the lights are first switched on at 09:00 A.M. Sharp, at what time will they change simultaneously? | CO5 | K1 |
| | (a) 9: 08: 04 (b) 9: 08: 24 (c) 9: 08: 44 (d) None of these | | |
| 52. | Three runners running around a circular track can complete one revolution in 2, 4, and 5.5 hours respectively, When will they meet at the starting point? | CO5 | K1 |
| | (a) 22 (b) 33 (c) 11 (d) 44 | | |
| 53. | The number A is not divisible by 3. Which of the following will not be divisible by 3? | CO5 | K1 |
| | (a) 9 x A (b) 2 x A (c) 18 x A (d) 24 x A | | |
| | Find the highest common factor of 38 and 39. | CO5 | K1 |
| 54. | (a) 1 (b) 3 (c) 13 (d) 19 | | |
| 55. | The L.C.M. of two numbers is 96. The numbers are in the ratio 3:4. Then, the larger number is: | CO5 | K1 |
| | (a) 28 (b) 32 (c) 40 (d) 64 | | |
| 56. | In the first 15 overs of a cricket game, the run rate was only 5.4. What should be the run rate in the remaining 35 overs to reach the target of 291 runs? | CO6 | K3 |
| | (a) 6 (b) 6.25 (c) 6.5 (d) 6.75 | | |
| 57. | The average of 6 observations is 12. The average decreases by 1 when the new observation is included. What is the new observation? | CO6 | K3 |
| | (a) 1 (b) 3 (c) 5 (d) 6 | | |

58. The average score of Rahul, Manish and Suresh are 63. Rahul's score is 15 less than Ajay and 10 more than Manish. If Ajay scored 30 marks more than the average score of Rahul, Manish, and Suresh, then what is the sum of Manish's and Suresh's scores?
(a) 120 (b) 111 (c) 117 (d) Cannot be determined
59. If the average of five numbers is 80, and three of the numbers are 130, 140, and 50, what are the remaining two numbers?
(a) 45, 55 (b) 50, 60 (c) 35, 45 (d) 40, 50
60. The total marks obtained by a student in three subjects A, B, and C together is 80 more than the marks obtained by B. What are the average marks obtained by A and C together?
(a) 42 (b) 38 (c) 40 (d) 56
61. Find the sum of all numbers divisible by 6 in between 100 to 400.
(a) 12000 (b) 2400 (c) 12450 (d) 12445
62. There is an AP 1, 3, 5.... Which term of this AP is 55?
(a) 25th (b) 26th (c) 27th (d) 28th
63. Find the first term of the AP series in which the 10th term is 6 and the 18th term is 70.
(a) 76 (b) -76 (c) 66 (d) -66
64. Find the nth term of the series 3, 7, 11, 15,...
(a) $2(2n+1)$ (b) $5n + 2$ (c) $5n - 2$ (d) $4n - 1$
65. A man receives Rs. 60 for the first week and Rs. 3 more each week than the preceding week. How much does he earn by the 20th week?
(a) Rs. 1770 (b) Rs. 1620 (c) Rs. 1890 (d) Rs. 1790
66. Which of the following equations has 2 as a root?
(a) $x^2 - 4x + 5 = 0$ (b) $x^2 + 3x - 12 = 0$ (c) $2x^2 - 7x + 6 = 0$ (d) $3x^2 - 6x - 2 = 0$
67. The sum of the roots of the quadratic equation $3x^2 - 9x + 5 = 0$ is?
(a) 3 (b) 6 (c) -3 (d) 2
68. Given equation is the quadratic equation?
(a) $(5/x) - 3 = x^2$ (b) $x(x + 5) = 2$ (c) $n - 1 = 2n$ (d) $(1/x^2) + (x + 2) = x$
69. The maximum number of roots for a quadratic equation is equal to
(a) 0 (b) 1 (c) 2 (d) 3
70. The equation of $(x+1)^2 - x^2 = 0$ has the number of real roots equal to:
(a) 1 (b) 2 (c) 3 (d) 4
71. Given interchange: sign '+' and '-' and numbers 5 and 8. Which of the following is correct?
(a) $82 - 35 + 55 = 2$ (b) $82 - 35 + 55 = 102$ (c) $85 - 38 + 85 = 132$ (d) $52 - 35 + 55 = 72$
72. Select the correct combination of mathematical signs to replace the '*' sign to balance the equation. $9 * 4 * 22 * 14$
(a) $x = -$ (b) $\times - =$ (c) $= - \times$ (d) $- \times =$
73. If '+' means ' $\times$ ', '-' means ','; ' $\times$ ' means '-' and ',' means '+' then $9 + 8, 8 - 4 \times 9 = ?$
(a) 26 (b) 17 (c) 65 (d) 11

74. If '+' means '×', '-' means '÷', '÷' means '+', and '×' means '-'; then what will be the value of the following expression?
 $540 - 36 + 12 \div 75 \times 55$
 (a) 255 (b) 512 (c) 180 (d) None of these
75. Given interchanges:
 Signs + and x and numbers 4 and 5.
 (a) $5 \times 4 + 20 = 40$ (b) $5 \times 4 + 20 = 85$ (c) $5 \times 4 + 20 = 104$ (d) $5 \times 4 + 20 = 95$
76. A is B's sister. C is B's mother. D is C's father. E is D's mother. Then, how is A related to D?
 (a) Grandfather (b) Grandmother (c) Daughter (d) Granddaughter
77. A woman introduces a man as the son of the brother of her mother. How is the man, related to the woman?
 (a) Nephew (b) Son (c) Cousin (d) Uncle to Grandson
78. If M is the brother of N; B is the brother of N, and M is the brother of D, then which of the following statements is true?
 (a) N is the brother of B (b) N is the brother of D
 (c) M is the brother of B (d) D is the brother of M
79. Amit and Ritu are a couple. Mrinal and Shonal are brothers. Mrinal is the brother of Ritu. How is Shonal related to Amit?
 (a) Brother-in-Law (b) Brother (c) Cousin (d) Uncle
80. A is the brother of B. C is the father of A. D is the brother of E. E is the daughter of B. Then, the uncle of D is:
 (a) A (b) B (c) C (d) E
81. In a row, all the persons are facing north. Sarah is 24th from the left end, and on the right side of Sarah, there are only 17 persons. Find out the total number of persons in this queue.
 (a) 40 (b) 41 (c) 39 (d) None of these
82. In a batch of 100 students, Kavya's rank is 50th from the top and Lucky's rank is 68th from the bottom. How many students are there between Kavya and Lucky (if no two persons have the same rank)?
 (a) Less than 10 (b) 16 (c) CND (d) 13
83. Sameer is 20th from the left end of a row and Arun is 12th from the right end of a row. If they interchanged their positions, then Arun's ranks become 10 from the right end. Find the total number of persons in the row.
 (a) 28 (b) 29 (c) 30 (d) None of these
84. In a row all the persons are facing north, Rahul is 33rd from the left end, and on the right side of Rahul, there are only 16 persons. Find out the total number of people in this queue.
 (a) 49 (b) 50 (c) 51 (d) Cannot be determined
85. In a row of boys, Karthick is seventh from the left and Suganya is twelfth from the right. If they interchange their positions, Karthick becomes twenty second from the left. How many boys are there in the row?
 (a) 19 (b) 29 (c) 31 (d) 33

- 86 Ajay walks 24 km towards East and turns to right hand side and takes a drive of another 10 km. He then turning to his right (drives towards West) another 10 km. He then turns to his left & walks another 8 km. After that, he turns to his right & travels 14 km. How far is he from his initial point & in which direction?
(a) 20 km East (b) 18 km south (c) 16 km West (d) 10 km South
- 87 Sandeep walks 60m to the east, then he turns left and walks for 50 m, then turns right and went 70 m and then turns right again and went 50 m. How far was Sandeep from the starting point?
(a) 90 m (b) 70 m (c) 50 m (d) 130 m
- 88 A boy rode his bicycle Northward, then turned left and rode 1 km and again turned left and rode 2 km. He found himself 1 km west of his starting point. How far did he ride northward initially?
(a) 1 km (b) 2 km (c) 3 km (d) 5 km
- 89 A man is facing north. He turns 45 degrees in the clockwise direction then another 180 degrees in the same direction and then 45 degrees in the anticlockwise direction. Find which direction he is facing now.
(a) North (b) East (c) West (d) South
- 90 Ahmedabad is to the southwest of Bangalore, Chennai is to the east of Ahmedabad and southeast of Bangalore and Delhi is to the north of Chennai in line with Ahmedabad - Bangalore. In which direction of Bangalore is Delhi located?
(a) South (b) South - West (c) North (d) North - East

Directions (91-95): In each question below are given two statements followed by two conclusions numbered I and II. You have to take the given two statements to be true even if they seem to be at variance from commonly known facts. Read the conclusion and then decide which of the given conclusions logically follows from the two given statements, disregarding commonly known facts.

- 91 **Statements:** Some Cats are Rats. All bats are tables. All Rats are Bats.
Conclusion: I. Some Cats are bats II. All bats are rats III. All tables are cats IV. All bats are cats
(a) Only I & II follow (b) Only II follows
(c) Only I & IV follow (d) None of these
- 92 **Statements:** All Carrots are birds. Some telephones are Carrots. All bedsheets are telephone.
Conclusion: I. All bedsheets are birds II. Some bedsheets are birds III. Some birds are telephone IV. All telephones are birds
(a) Only I follow (b) Only II follows
(c) Only I and III follow (d) Only III follows
- 93 **Statements:** All the harmoniums are instruments. All the instruments are flutes.
Conclusion: All the flutes are instruments. All the harmoniums are flutes.
(a) Only conclusion I follows (b) Only conclusion II follows
(c) Either I or II follows (d) Neither I nor II follows
- 94 **Statements:** All liquids are solids. Some solids are gases. All gases are clouds
Conclusion: Some clouds are solids Some clouds are liquids Some gases are liquids Some solids are clouds
(a) Only conclusion I follows (b) Only conclusions I and IV follow
(c) Either I or II follows (d) Neither I nor II follows

95. **Statements:** All roads are water. Some waters are boats. CO5 K2
Conclusions: Some boats are roads. All waters are boats.
 (a) Only conclusion I follows (b) Only conclusions I and IV follow
 (c) Either I or II follows (d) Neither I nor II follows

Directions (96-100): In these questions, the relationship between different elements is shown in the statements. Some statements are followed by some conclusions. Choose the correct answer based on the information given below.

96. **Statements:** $B > A \geq T > F = Y \leq S < D$ CO5 K2
Conclusions: $F < D, A > S$
 (a) Only conclusion I follows (b) Only conclusion II follows
 (c) Either conclusion I or conclusion II follows (d) Both conclusions follow
 (e) Neither Conclusion I nor Conclusion II follows
97. **Statements:** $Y < O \leq G \leq K = U > L > P$ CO5 K2
Conclusions: $O = U, U > O$
 (a) Only conclusion I follows (b) Only conclusion II follows
 (c) Either conclusion I or conclusion II follows (d) Both conclusions follow
 (e) Neither Conclusion I nor Conclusion II follows
98. **Statements:** $M < T < G \leq J = U > Y > R$ CO5 K2
Conclusions: $G < U, J > R$
 (a) Only conclusion I follows (b) Only conclusion II follows
 (c) Either conclusion I or conclusion II follows (d) Both conclusions follow
 (e) Neither Conclusion I nor Conclusion II follows
99. **Statements:** $P \leq R \leq C = S > Q > T$ CO5 K2
Conclusions: I. $P < Q$ II. $S \geq P$
 (a) Only conclusion I follows (b) Only conclusion II follows
 (c) Either conclusion I or conclusion II follows (d) Both conclusions follow
 (e) Neither Conclusion I nor Conclusion II follows
100. **Statements:** $T > D \geq P, F \geq P = R$ CO5 K2
Conclusions: $T > R, D > F$
 (a) Only conclusion I follows (b) Only conclusion II follows
 (c) Either conclusion I or conclusion II follows (d) Both conclusions follow
 (e) Neither Conclusion I nor Conclusion II follows

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